

Headline metrics — std log-IV space (overall, n_test=1007) (pred_len=21)

display	n_seeds	RMSE ($\pm \sigma_{\text{seed}}$) [95% CI]	MSE	MAE	R ² vs pers [95% CI]	var ratio	params	val_loss
dlinear_ens	6 (ens)	0.4672 [0.3201, 0.6583]	0.2183	0.2890	+0.0530 [-0.0335, +0.1889]	0.854	—	—
dlinear	6	0.4709 $\pm$ 0.0053 [0.3253, 0.6609]	0.2217	0.2936	+0.0382 [-0.0531, +0.1623]	0.854	—	—
patchtst_ens	6 (ens)	0.4741 [0.3185, 0.6782]	0.2248	0.2913	+0.0249 [-0.0745, +0.2003]	0.836	—	—
persistence	—	0.4801 [0.3359, 0.6659]	0.2305	0.3001	— (reference)	0.996	—	—
patchtst	6	0.4840 $\pm$ 0.0080 [0.3294, 0.6856]	0.2342	0.3012	-0.0162 [-0.1128, +0.1319]	0.836	—	—
tucker_dlinear_ens	6 (ens)	0.4939 [0.3369, 0.6961]	0.2439	0.3082	-0.0582 [-0.1875, +0.1131]	0.759	—	—
gwn_ens	6 (ens)	0.4963 [0.3366, 0.7078]	0.2464	0.3063	-0.0687 [-0.1798, +0.0928]	0.887	—	—
tucker_dlinear	6	0.5002 $\pm$ 0.0129 [0.3446, 0.7027]	0.2502	0.3146	-0.0855 [-0.2218, +0.0722]	0.759	—	—
var	1	0.5043 [0.3265, 0.7301]	0.2544	0.2976	-0.1035 [-0.2450, +0.1545]	0.828	—	—
gwn	6	0.5137 $\pm$ 0.0198 [0.3602, 0.7198]	0.2639	0.3234	-0.1447 [-0.2572, -0.0224]	0.887	—	—
hot/kronecker_sum	1	0.5170 [0.3548, 0.7273]	0.2673	0.3292	-0.1597 [-0.3220, -0.0068]	0.887	84,949	0.1751
hot/kronecker_product	1	0.5306 [0.3465, 0.7585]	0.2816	0.3287	-0.2215 [-0.4529, +0.0315]	0.705	167,765	0.1708

Headline metrics — COVID (n_test=254) (pred_len=21)

display	n_seeds	RMSE (± σ_seed) [95% CI]	MSE	MAE	R² vs pers [95% CI]	var ratio	params	val_loss
dlinear_ens	6 (ens)	0.7277 [0.3303, 1.1496]	0.5295	0.4009	+0.0782 [-0.0258, +0.3923]	0.854	—	—
dlinear	6	0.7328 ± 0.0043 [0.3424, 1.1518]	0.5370	0.4106	+0.0651 [-0.0395, +0.3542]	0.902	—	—
patchtst_ens	6 (ens)	0.7529 [0.3349, 1.1951]	0.5669	0.4136	+0.0130 [-0.1132, +0.3799]	0.836	—	—
persistence	—	0.7579 [0.3898, 1.1661]	0.5744	0.4455	— (reference)	1.043	—	—
patchtst	6	0.7704 ± 0.0125 [0.3659, 1.2116]	0.5935	0.4383	-0.0332 [-0.1634, +0.2522]	0.889	—	—
tucker_dlinear_ens	6 (ens)	0.7818 [0.3846, 1.2229]	0.6112	0.4557	-0.0641 [-0.2529, +0.2359]	0.759	—	—
gwn_ens	6 (ens)	0.7886 [0.3575, 1.2496]	0.6219	0.4369	-0.0827 [-0.2259, +0.2585]	0.887	—	—
tucker_dlinear	6	0.7898 ± 0.0237 [0.3954, 1.2311]	0.6239	0.4652	-0.0861 [-0.2927, +0.1808]	0.789	—	—
gwn	6	0.8144 ± 0.0350 [0.4156, 1.2678]	0.6632	0.4707	-0.1547 [-0.3172, +0.0542]	0.920	—	—
hot/kronecker_sum	1	0.8149 [0.3732, 1.2718]	0.6641	0.4731	-0.1562 [-0.4074, +0.1755]	0.863	84,949	0.1751
var	1	0.8342 [0.3843, 1.3063]	0.6959	0.4640	-0.2115 [-0.4223, +0.2001]	0.883	—	—
hot/kronecker_product	1	0.8665 [0.4057, 1.3450]	0.7508	0.5080	-0.3071 [-0.7055, +0.0782]	0.704	167,765	0.1708

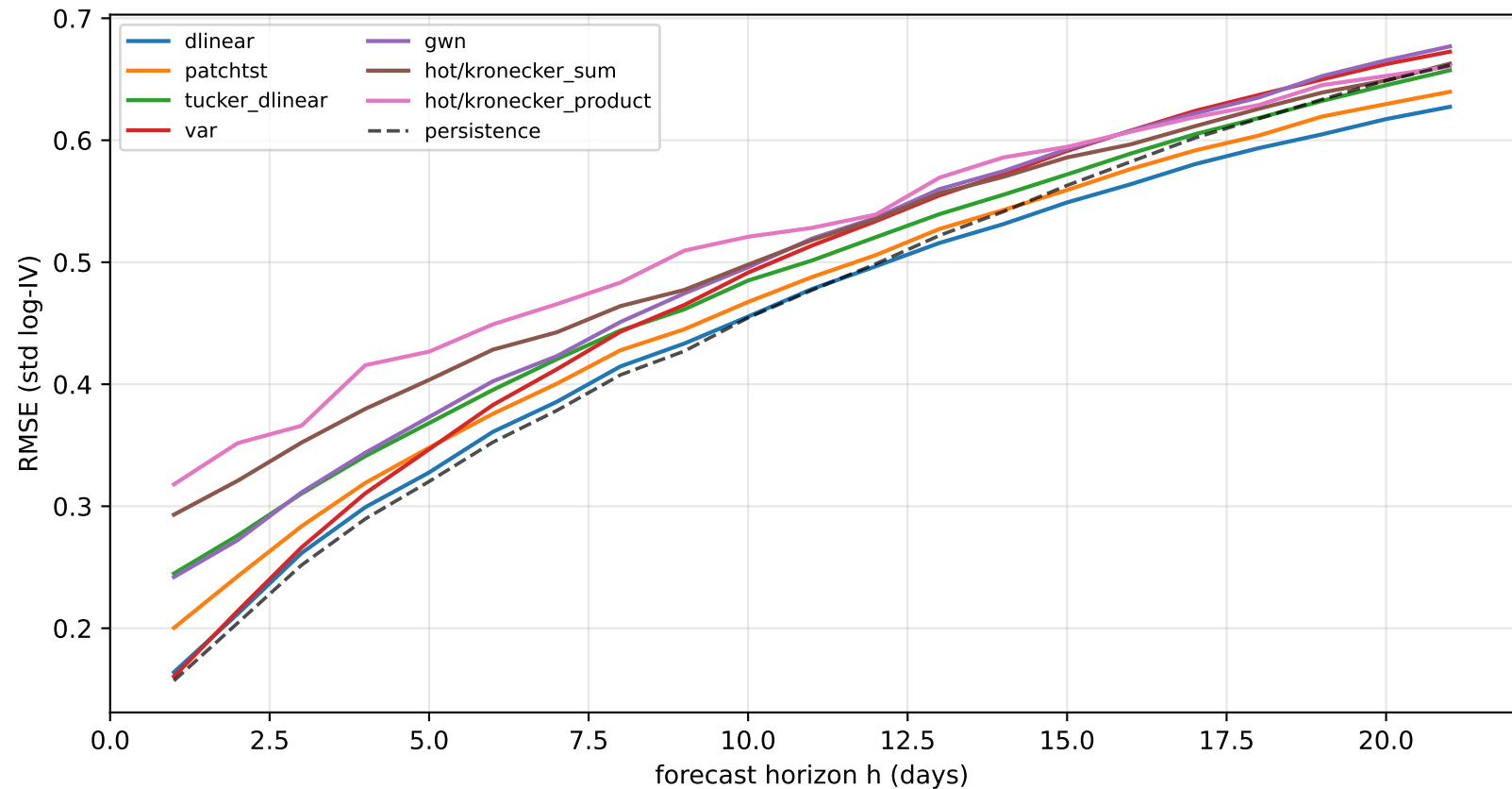
display	n_seeds	RMSE (± σ_{seed}) [95% CI]	MSE	MAE	R ² vs pers [95% CI]	var ratio	params	val_loss
var	1	0.2739 [0.2278, 0.3066]	0.0750	0.2025	+0.2194 [+0.1090, +0.3619]	0.805	—	—
tucker_dlinear_ens	6 (ens)	0.2794 [0.2319, 0.3141]	0.0781	0.2059	+0.1877 [+0.0116, +0.3539]	0.759	—	—
patchtst_ens	6 (ens)	0.2847 [0.2380, 0.3178]	0.0810	0.2098	+0.1569 [+0.0683, +0.2527]	0.836	—	—
tucker_dlinear	6	0.2867 ± 0.0147 [0.2410, 0.3205]	0.0822	0.2126	+0.1447 [-0.0433, +0.3134]	0.725	—	—
patchtst	6	0.2900 ± 0.0062 [0.2439, 0.3224]	0.0841	0.2147	+0.1251 [+0.0275, +0.2230]	0.818	—	—
dlinear_ens	6 (ens)	0.2933 [0.2451, 0.3280]	0.0860	0.2173	+0.1049 [-0.0419, +0.2401]	0.854	—	—
dlinear	6	0.2957 ± 0.0044 [0.2478, 0.3297]	0.0875	0.2194	+0.0902 [-0.0635, +0.2263]	0.838	—	—
hot/kronecker_product	1	0.3034 [0.2500, 0.3435]	0.0920	0.2270	+0.0427 [-0.1527, +0.2348]	0.715	167,765	0.1708
persistence	—	0.3100 [0.2611, 0.3477]	0.0961	0.2246	— (reference)	0.997	—	—
gwn_ens	6 (ens)	0.3144 [0.2563, 0.3556]	0.0988	0.2331	-0.0283 [-0.1887, +0.1274]	0.887	—	—
hot/kronecker_sum	1	0.3238 [0.2722, 0.3632]	0.1049	0.2417	-0.0909 [-0.1836, -0.0060]	0.882	84,949	0.1751
gwn	6	0.3259 ± 0.0255 [0.2696, 0.3655]	0.1062	0.2432	-0.1048 [-0.2705, +0.0460]	0.899	—	—

display	n_seeds	RMSE (± σ_seed) [95% CI]	MSE	MAE	R² vs pers [95% CI]	var ratio	params	val_loss
persistence	—	0.3884 [0.3254, 0.4580]	0.1509	0.2936	— (reference)	1.038	—	—
var	1	0.3923 [0.3194, 0.4751]	0.1539	0.2921	-0.0199 [-0.3387, +0.2123]	0.865	—	—
patchtst_ens	6 (ens)	0.3999 [0.3318, 0.4756]	0.1599	0.3041	-0.0597 [-0.3485, +0.1440]	0.836	—	—
dlinear_ens	6 (ens)	0.4051 [0.3319, 0.4817]	0.1641	0.3032	-0.0876 [-0.3881, +0.1370]	0.854	—	—
patchtst	6	0.4066 ± 0.0233 [0.3412, 0.4806]	0.1653	0.3113	-0.0959 [-0.3905, +0.1044]	0.883	—	—
dlinear	6	0.4088 ± 0.0134 [0.3374, 0.4847]	0.1671	0.3076	-0.1077 [-0.4112, +0.1140]	0.895	—	—
gwn_ens	6 (ens)	0.4136 [0.3530, 0.4817]	0.1711	0.3172	-0.1340 [-0.3542, +0.0246]	0.887	—	—
hot/kronecker_sum	1	0.4271 [0.3602, 0.4995]	0.1824	0.3355	-0.2088 [-0.4405, +0.0045]	1.104	84,949	0.1751
gwn	6	0.4312 ± 0.0250 [0.3784, 0.4943]	0.1859	0.3335	-0.2322 [-0.4979, -0.0519]	1.053	—	—
hot/kronecker_product	1	0.4356 [0.3571, 0.5235]	0.1897	0.3339	-0.2576 [-0.7106, +0.0577]	0.794	167,765	0.1708
tucker_dlinear_ens	6 (ens)	0.4443 [0.3650, 0.5343]	0.1974	0.3376	-0.3083 [-0.7546, +0.0271]	0.759	—	—
tucker_dlinear	6	0.4501 ± 0.0200 [0.3717, 0.5397]	0.2026	0.3429	-0.3430 [-0.7910, -0.0073]	0.818	—	—

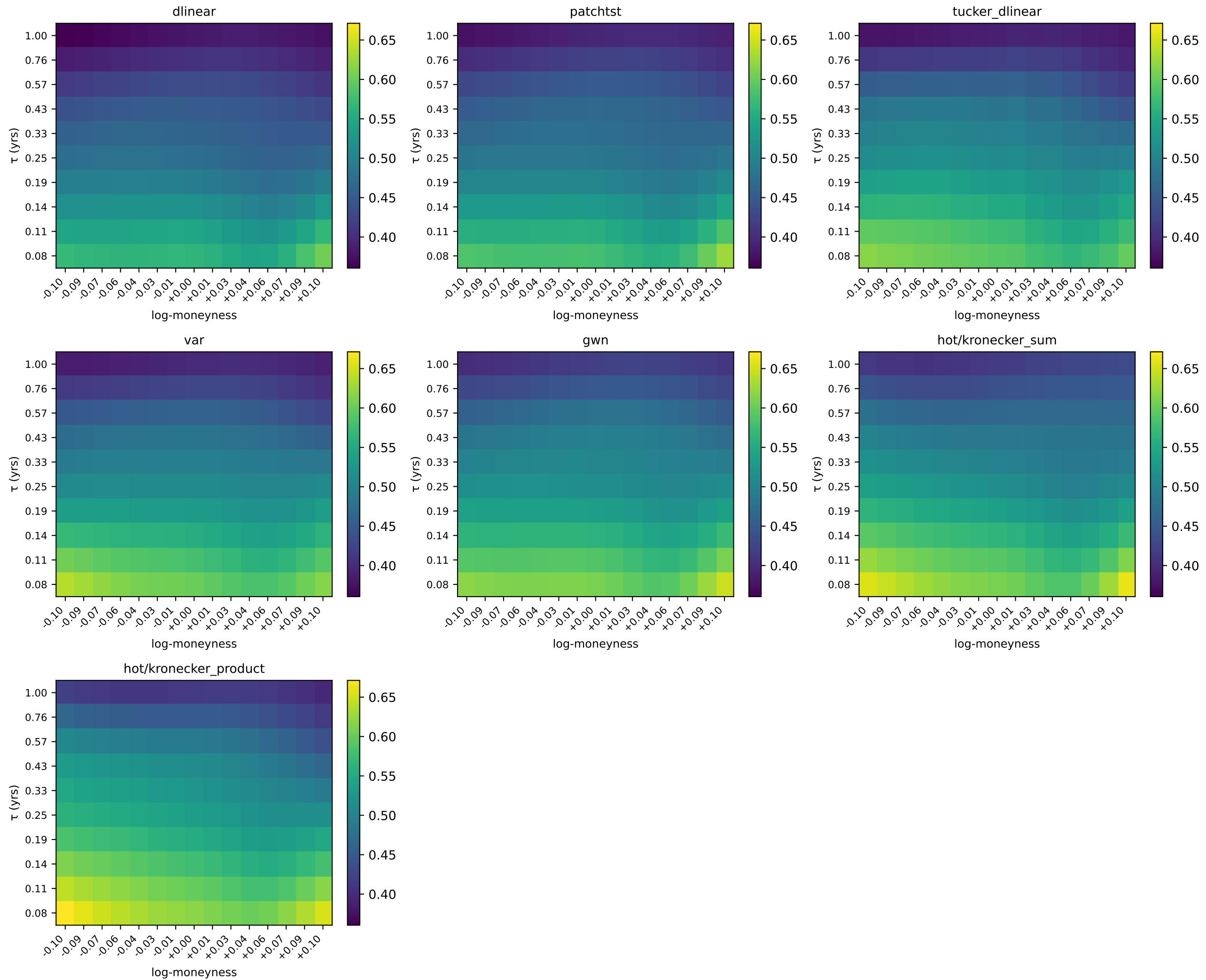
Headline metrics — Normalisation (n_test=250) (pred_len=21)

display	n_seeds	RMSE (± σ_seed) [95% CI]	MSE	MAE	R ² vs pers [95% CI]	var ratio	params	val_loss
tucker_dlinear_ens	6 (ens)	0.2910 [0.2492, 0.3417]	0.0847	0.2318	+0.1229 [-0.0715, +0.3089]	0.759	—	—
patchtst_ens	6 (ens)	0.2953 [0.2553, 0.3452]	0.0872	0.2361	+0.0967 [-0.0615, +0.2503]	0.836	—	—
var	1	0.2957 [0.2627, 0.3424]	0.0874	0.2300	+0.0943 [-0.0989, +0.2337]	0.935	—	—
tucker_dlinear	6	0.2963 ± 0.0054 [0.2549, 0.3458]	0.0878	0.2359	+0.0909 [-0.1079, +0.2766]	0.805	—	—
hot/kronecker_product	1	0.2968 [0.2603, 0.3383]	0.0881	0.2440	+0.0878 [-0.1438, +0.2787]	0.856	167,765	0.1708
gwn_ens	6 (ens)	0.2984 [0.2578, 0.3484]	0.0891	0.2365	+0.0775 [-0.1849, +0.2893]	0.887	—	—
patchtst	6	0.2996 ± 0.0039 [0.2603, 0.3484]	0.0898	0.2391	+0.0702 [-0.0880, +0.2245]	0.858	—	—
dlinear_ens	6 (ens)	0.2997 [0.2523, 0.3652]	0.0898	0.2332	+0.0695 [-0.0731, +0.2489]	0.854	—	—
dlinear	6	0.3025 ± 0.0025 [0.2552, 0.3669]	0.0915	0.2353	+0.0523 [-0.0864, +0.2269]	0.879	—	—
gwn	6	0.3087 ± 0.0109 [0.2696, 0.3570]	0.0953	0.2442	+0.0130 [-0.2582, +0.2374]	0.910	—	—
persistence	—	0.3107 [0.2681, 0.3708]	0.0965	0.2351	— (reference)	1.034	—	—
hot/kronecker_sum	1	0.3366 [0.2908, 0.3967]	0.1133	0.2647	-0.1734 [-0.5127, +0.0837]	1.031	84,949	0.1751

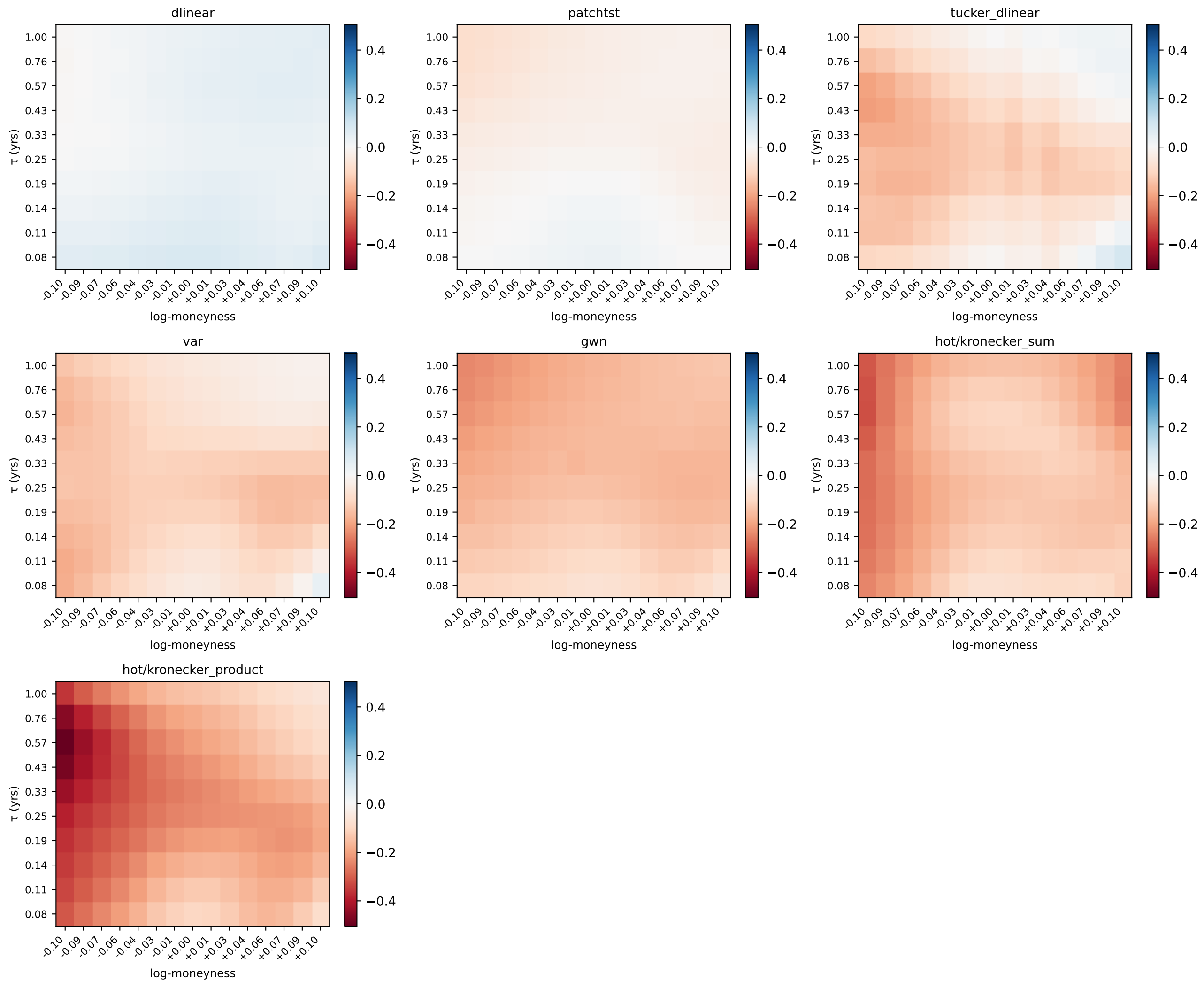
RMSE vs forecast horizon



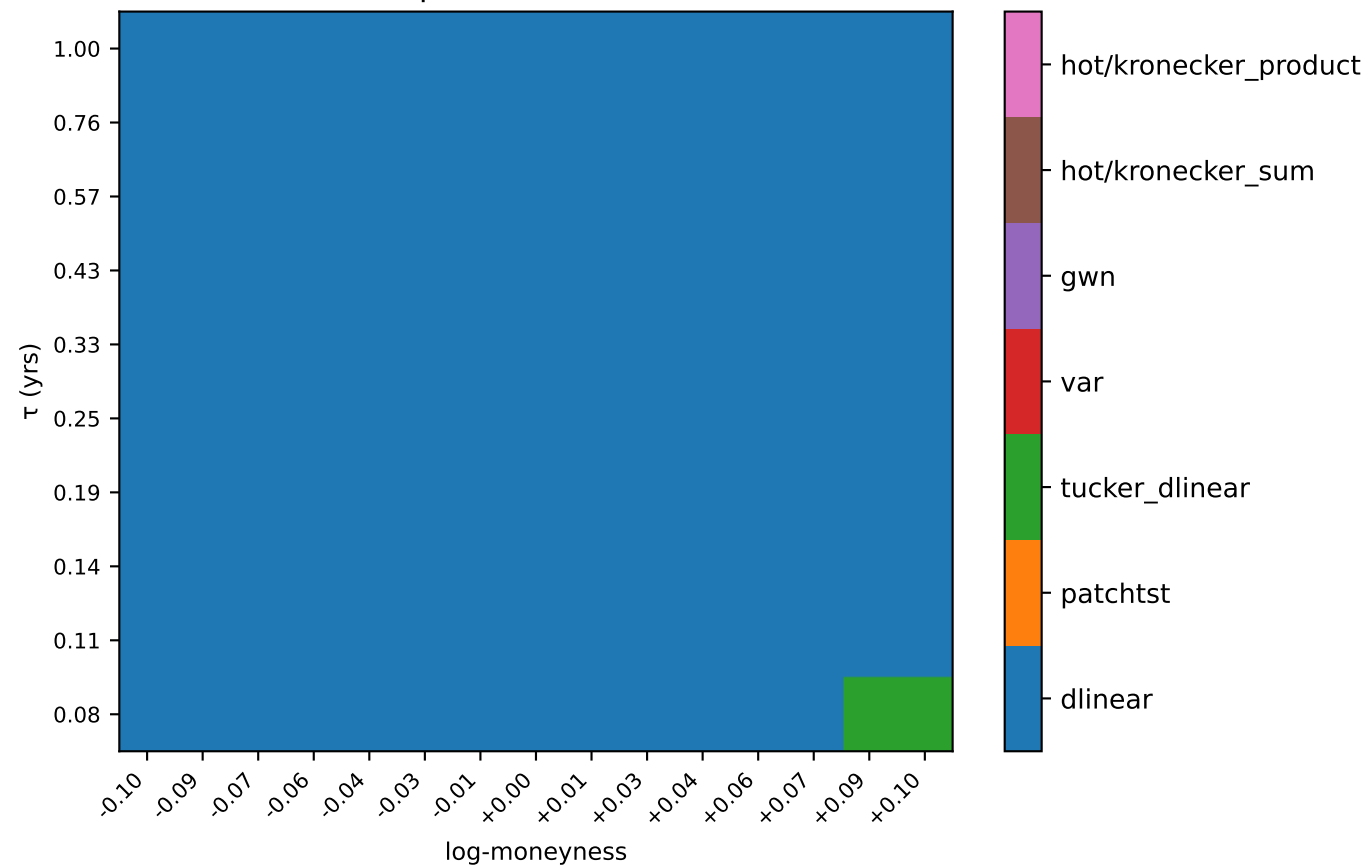
Per-cell RMSE (std log-IV) — each panel is one model

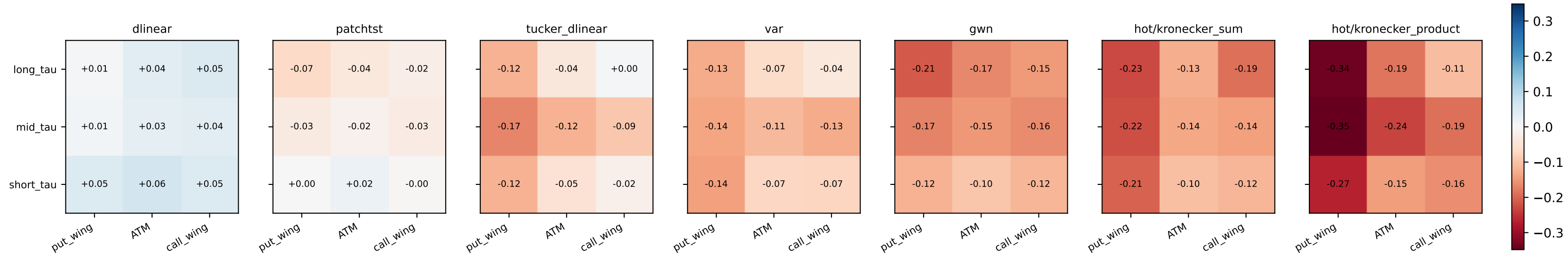


R^2 vs persistence per cell (red = worse than persistence)

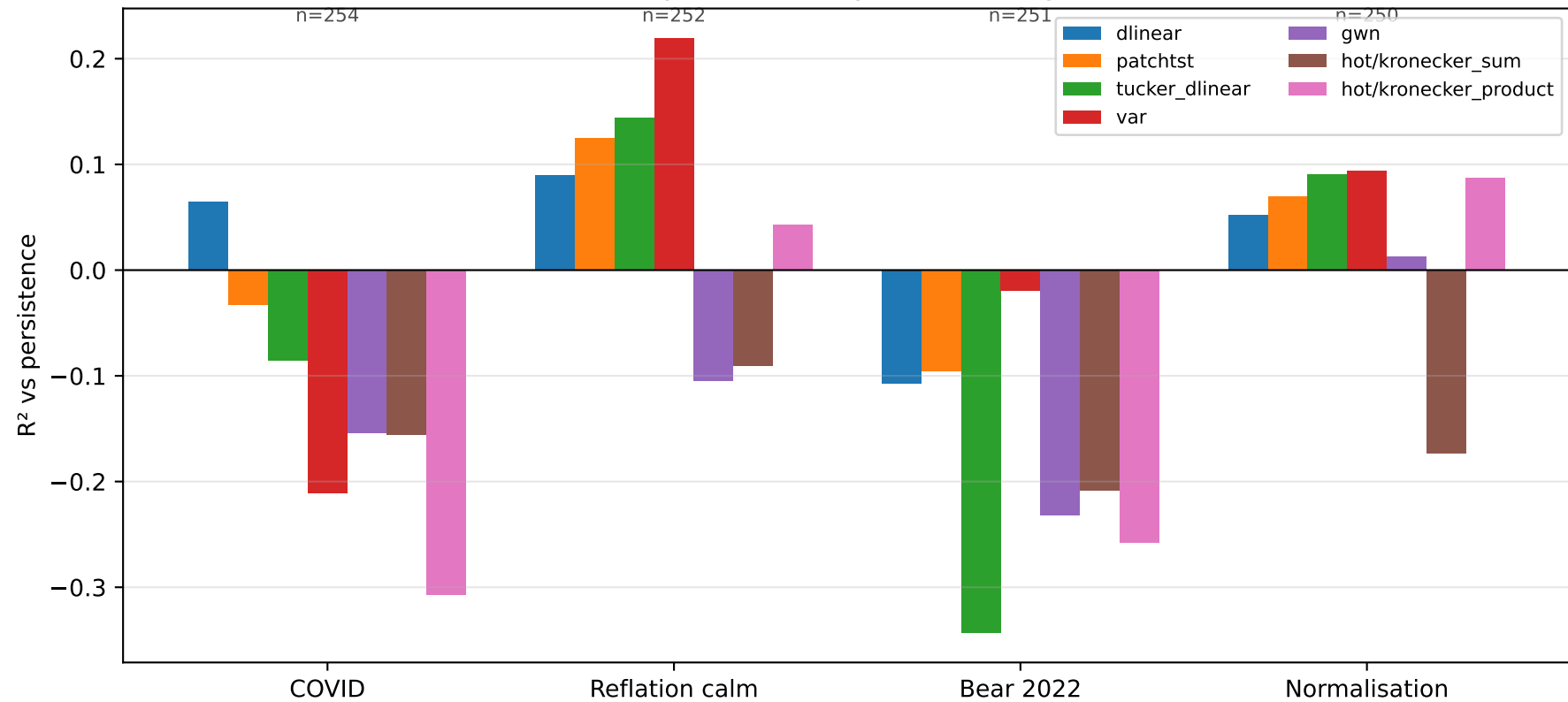


Winner per cell (lowest RMSE)

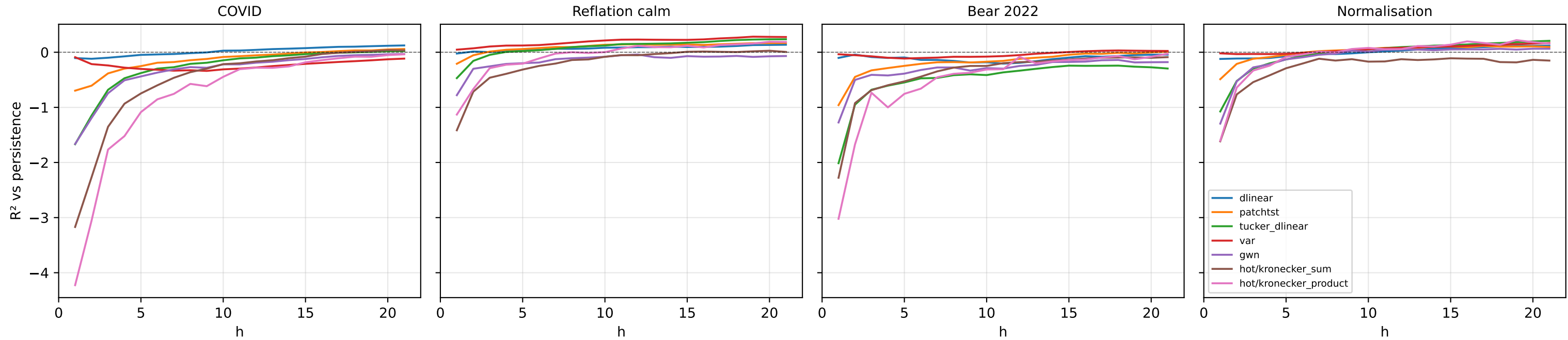


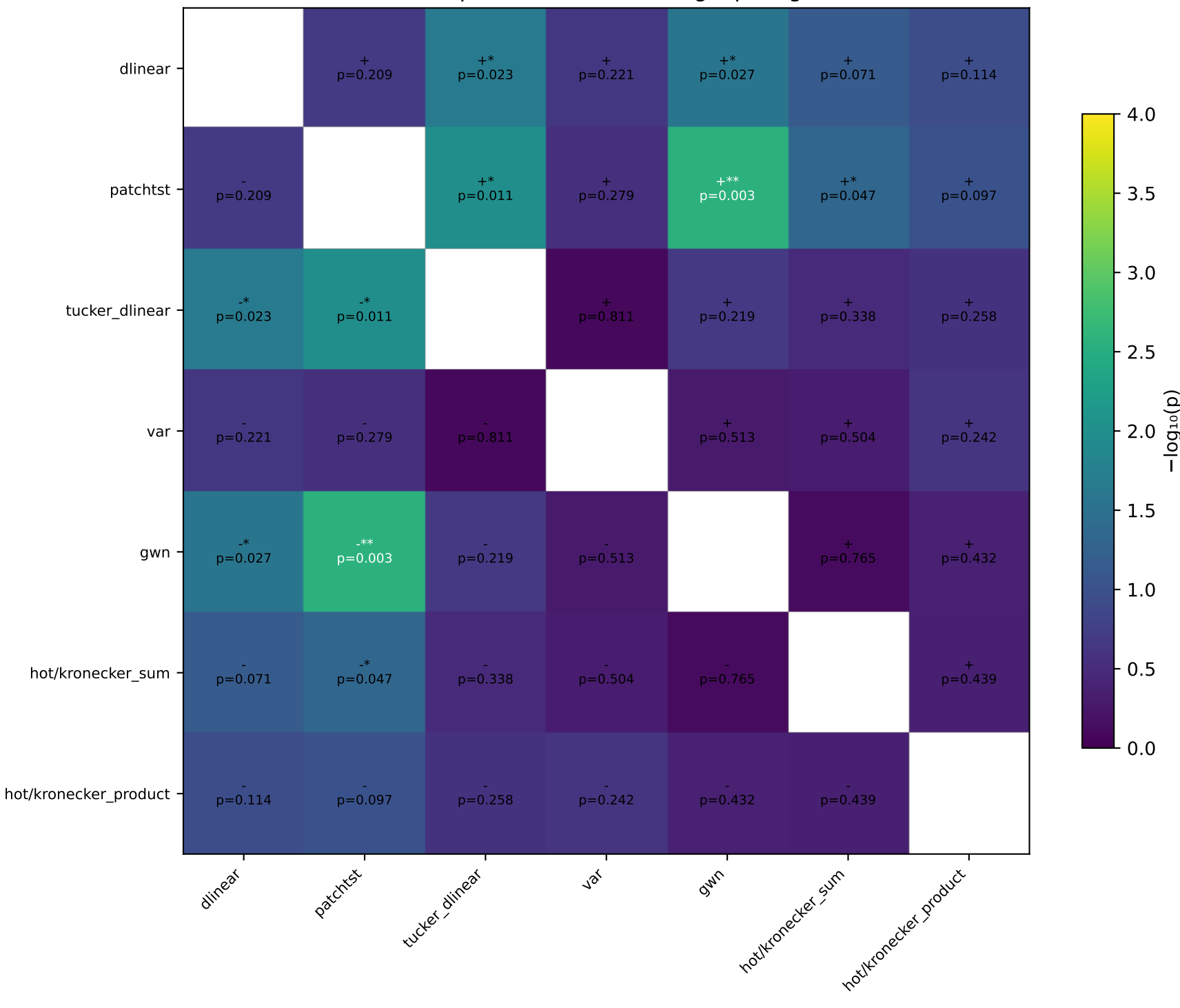
R² vs persistence by surface zone

R² vs persistence by calendar regime

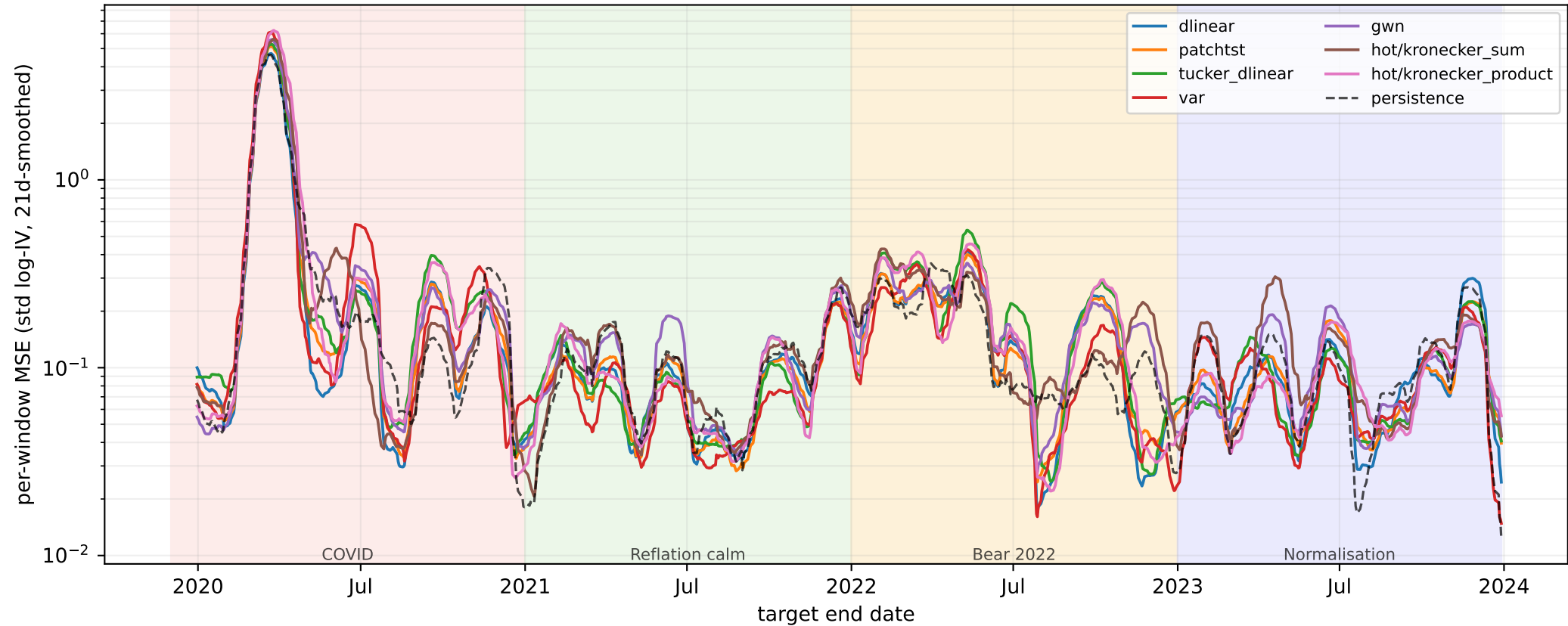


R^2 vs persistence by horizon, faceted by regime

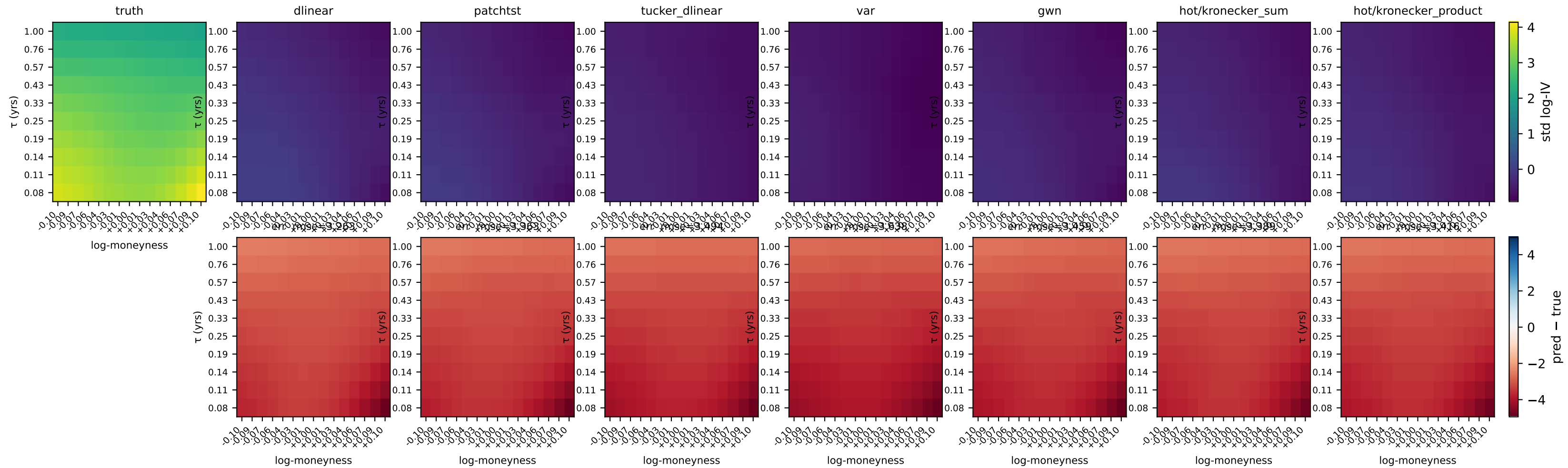


Diebold-Mariano pairwise — color: $-\log_{10}(p)$, sign: row vs col

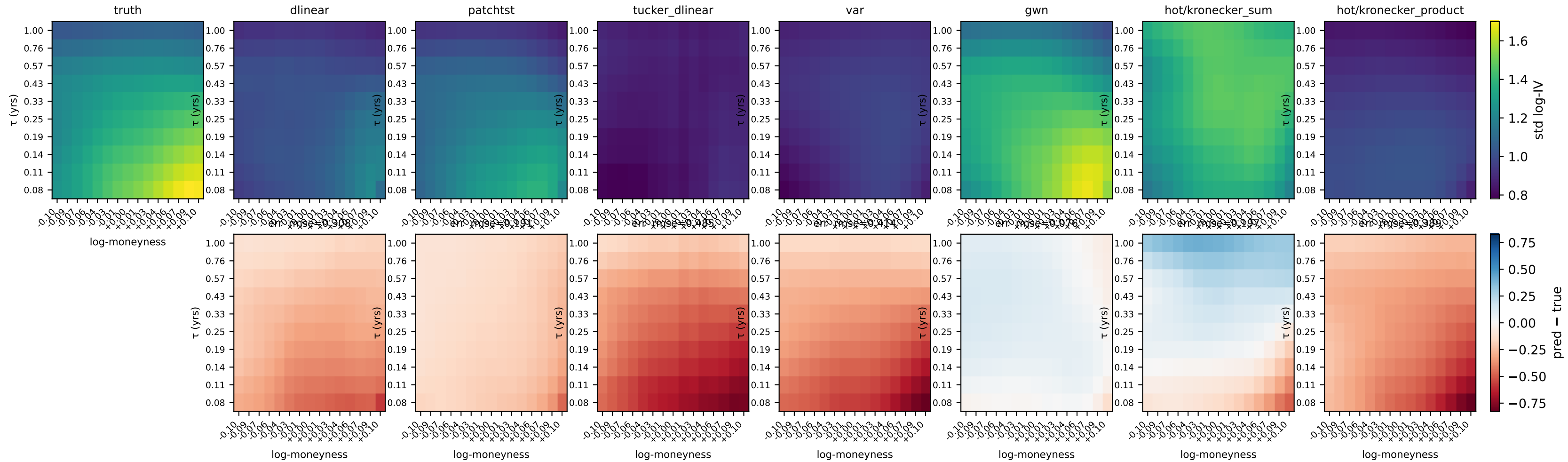
Per-window MSE over time, regime-shaded



Example forecast — covid: target 2020-03-23 (h=21)



Example forecast — bear: target 2022-06-15 (h=21)



Example forecast — calm: target 2023-08-15 (h=21)

